Problem Set VIII

Political Science 43401
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- 5.1 (p 284)
- 5.8 (p 290)
- 9.20 (p 518)
- 9.22 (a) (b) (p 519)
- 9.26 (p 519)
- 9.27 (p 519)
- 9.30 (p 520)
- Suppose $\bar{X}$ is the mean of 100 observations from a population with mean μ and variance σ^2. Find limits between which $\bar{X} - \mu$ will lie with a probability of at least .85 using the Central Limit Theorem.
- Consider the function $f(x, y)$ defined by the following table

	$y = 1$	2	3
$x = 1$	.1	0	.1
2	0	.2	.2
3	.1	.2	.1

1. Verify that $f(x, y)$ is a PDF.
2. Find the marginal probability functions of X and Y.
3. Compute E(X) and V(X).
4. Compute the covariance between X and Y.
5. Find $P(X < Y)$.
6. Are X and Y independent? Justify your answer.

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